INCH-POUND

MIL-H-18158E(SH)

30 November 1990
SUPERSEDING
MIL-H-18158D(SH)
21 October 1980

MILITARY SPECIFICATION

HOSE DUCTING, LOW PRESSURE, NAVAL SHIPBOARD PORTABLE, VENTILATING SET USE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lightweight, cloth covered, wire reinforced, helical seam, portable ventilation hose used aboard Navy ships.

1.2 Classification. Hose assembly and the stowage rack shall be of the following classes, as specified (see 6.2).

Class 1 – Magnetic
Class 2 – Nonmagnetic

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4720

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:01Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-665	Boxes, Paperboard, Metal Edged and Components
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-15145	Enamel, Zinc Dust Pigmented, Fresh Water Tank Protective (Formula No. 102)
DOD-P-15328	Primer (Wash), Pretreatment, (Formula No. 117 for Metals); (Metric)
MIL-L-19140	Lumber and Plywood, Fire-Retardant Treated
DOD-P-21035	Paint, High Zinc Dust Content, Galvanizing Repair; (Metric)

STANDARDS

FEDERAL

FED-STD-162	Hose, Rubber, Visual Inspection Guide for
FED-STD-191	Textile Test Methods
FED-STD-751	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-278	Welding and Casting Standard
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods
MIL-STD-2073-1	DOD Material Procedures for Development and Application of Packaging Requirements

Source: https://assist.dla.mil -- Downloaded: 202610-18T11:01Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

803-607819 Hose 8, 10, 16 Inch and Pack for Portable Ventilation Set

(Application for copies should be addressed to: Commander, Portsmouth Naval Shipyard, Code 202.2, Portsmouth, NH 03801.)

2.2 **Non-Government publications.** The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

A 123	Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products; (DOD adopted)
A 525	Standard Specification for General Requirements for Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process; (DOD adopted)
A 569	Standard Specification for Steel, Carbon (0.15 Maximum, Percent), Hot-Rolled Sheet and Strip, Commercial Quality; (DOD adopted)
B 6	Standard Specification for Zinc; (DOD adopted)
B 159	Standard Specification for Phosphor Bronze Wire; (DOD adopted)
B 201	Standard Practice for Testing Chromate Coatings on Zinc and Cadmium Surfaces
B 209	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate; (DOD adopted)

Source: https://assist.dla.mil -- Downloaded: 202610-18T11:01Z
Check the source to verify that this is the current version before use.

B 633	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel; (DOD adopted)
A 764	Steel Wire, Carbon Drawn Galvanized, and Galvanized at Size for Mechanical Springs
D 3951	Standard Practice for Commercial Packaging; (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AIR MOVEMENT AND CONTROL ASSOCIATION (AMCS)

Standard 210	Laboratory Methods of Testing Fans for Rating

(Application for copies should be addressed to the Air Movement and Control Association, 30 West University Drive, Arlington Heights, IL 60004.)

AMERICAN WELDING SOCIETY (AWS)

A5.1	Covered Carbon Steel Arc Welding Electrodes

(Application for copies should be addressed to the American Welding Society, Inc., 500 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3 (see 6.4).

3.2 Material. Materials used in the construction of the hose assembly and hose storage rack shall be of a high grade and free from defects.

3.2.1 Dissimilar metals. Aluminum alloys shall not be placed in direct contact with copper or any copper bearing alloys.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:01Z
Check the source to verify that this is the current version before use.

3.2.2 **Recovered materials.** Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 **Construction.**

3.3.1 **Dimensions and tolerances.** The physical dimensions and tolerances of the hose assembly and the hose stowage rack shall conform to Drawing 803-607819.

3.3.1.1 **Hose assembly.** Each hose assembly, consisting of a length of hose to which required connectors are secured, shall be not less than 15 feet in length nor more than 15 feet 4 inches in length. The hose length shall be measured with the hose suspended vertically and with a 10-pound weight fastened to the lower end. Each length of hose shall be collapsible to a length not to exceed 30 inches.

3.3.2 **Hose.** The hose shall be of the helical seam type, stiffened with a helically wound spring wire. On stitched seam type the helical spring wire shall have a free length pitch substantially greater than the pitch of the seam, to exert axial tension on and to prevent transverse collapse of the hose. The pitch of the seam shall not exceed 4-1/4 inches and shall be not less than 3-1/4 inches.

3.3.3 **Strip.** A strip of the coated fabric (see 3.3.9), of the width necessary to give a pitch as specified in 3.3.2, shall be wound helically upon itself with overlapping edges between which shall be laid the helically wound spring wire.

3.3.4 **Hose end.** Each section of hose shall terminate at each end with a cuff which shall contain a retaining ring made from nylon rod. The hose end shall be collapsible by hand so as to be snapped within the hose connector. The hose end and retaining ring when not subjected to a deforming force shall return naturally to its original fully expanded shape.

3.3.5 **Ground.** Grounding straps shall be placed on each end of each hose section so as to ensure that a continuous ground will be established through the helical spring wire when hose sections are attached to end connectors.

3.3.6 **Helical spring.** The helical spring for holding the hose axially extended shall have a minimum wire diameter of 0.135 inch. The free length diameter of the coil, when collapsed to the length of the extended hose, shall be the same as the hose diameter. The ends of the helical spring shall be secured to the end connectors or hose end in such a manner as to form a continuous electrical ground for the helical spring and end connectors when multiple lengths of hose are joined together.

Source: https://assist.dla.mil -- Downloaded: 202640-18T11:01Z
Check the source to verify that this is the currentversion before use.

3.3.6.1 **Helical spring for class 1.** The helical spring for class 1 hose assembly shall be made of a galvanized hard drawn MB wire conforming to ASTM A 764 with the exception that it shall have a minimum tensile strength of 215,000 psi. The wire shall be zinc coated or rendered corrosion resistant by a nonphosphate process with a rust inhibiting agent.

3.3.6.2 **Helical spring for class 2.** The helical spring for class 2 hose assembly shall be of phosphor-bronze conforming to alloy A of ASTM B 159.

3.3.7 **Connectors.** Connectors furnished on hose lengths from all supply sources shall be interchangeable without the necessity of machining or handfitting of any kind. Each length of hose shall be supplied with the appropriate couplings for the connection of additional hose lengths or the portable blower. Connectors for both classes of hose assembly shall be of aluminum alloy in accordance with alloy 6061 T6 of ASTM B 209.

3.3.8 **Scuff strips.** The hose shall be provided with nonmetallic flame-resistant scuff strips in such a manner that the basic fabric and seams will be protected from excessive abrasion when dragged along a ship's deck or lowered through a hatch. The flame resistance of the scuff strip shall equal or exceed the minimum requirements of the fabric (see 3.3.9.3). Additionally, the scuff strip shall not wear through in less than 3,000 cycles (see 4.5.1).

3.3.9 **Fabric.**

3.3.9.1 **Base cloth.** The base cloth used shall be made of a polyester fiber that conforms to the requirements of this specification.

3.3.9.2 **Coating.** The compound used in coating the fabric shall be comprised of synthetic rubber utilizing polymerized chloroprene or chlorosulfonated polyethylene as the basic material. The coating on both sides of the fabric, and shall contain no material which will be injurious to the skin in handling.

3.3.9.3 **Coated fabric.** The coated fabric shall have the following characteristics:

a. Weight, ounces per square yard – 10 to 25

b. Breaking strength – pounds minimum, 175 warp, 140 fill

c. Abrasion resistance – wear through shall not occur in less than 7,500 cycles

d. Weathering resistance – breaking strength shall be not less than 75 percent of the initial strength.

e. Accelerated aging – breaking strength shall be not less than 75 percent of the initial strength

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:01Z
Check the source to verify that this is the current version before use.

g. Low temperature effect – shall not crack

h. Flame resistance –

(1) After flame, 2 seconds maximum

(2) Char length, 3 inches maximum (average)

(3) Melt and drip, the coated fabric shall not melt or drip when exposed to flame.

3.3.10 Thread. The thread used in all stitching shall be 3-ply nylon or polyester and shall exhibit a minimum average breaking strength of 8.50 pounds.

3.3.11 Seams. Seams shall conform to LSa-2 of FED-STD-751. The width of the seams shall contain the spring wire to form a satisfactory seam. Seams shall be stitched, cemented, or molded and cured to withstand the requirements and tests of this specification.

3.3.11.1 Seam stitching. Seams containing the spring wire shall be stitched with a two-needle sewing machine with the needles straddling the wire. Seams joining the end cuts to the hose body shall be stitched with a double-needle sewing machine; all other seams shall be sewn with a single- or double-needle machine. Stitching shall conform to type 301 of FED-STD-751. The space between rows of stitching and the edge strip shall be not less than 1/8 or more than 3/8 inch. Not more than two bobbin breaks shall be allowed in a 90-foot length of seam and where these occur the break shall be backstitched for a distance of not less than 1 inch.

3.4 Tensile strength. Hose shall withstand a tensile pull of at least 400 pounds before breaking.

3.5 Pressure drop. The air pressure drop through three standard lengths of hose shall not exceed 1.25 inch watergauge for the following size hoses at the specified air flow rate:

a. 8-inch hose at 500 cubic feet per minute (ft^3/min)
b. 10-inch hose at 850 ft^3/min
c. 16-inch hose at 2,000 ft^3/min.

The middle section of hose shall have a 90-degree bend and the air flow shall be measured downstream of the hose discharge (see 4.5.5).

3.6 Abrasion resistance, simulated service. The hose shall withstand a minimum of 200 abrasion cycles, simulating wear over a hatch coaming, before the reinforcing wire is exposed or holes are formed in fabric.

Source: https://assist.dla.mil -- Downloaded: 202610-18T11:01Z
Check the source to verify that this is the current version before use.

3.7 Branding. Each length of hose assembly shall be branded in a color contrasting to that of the hose, showing the manufacturer's name and address, date of manufacture by month and year, class of hose assembly and number of this specification.

3.8 Stowage racks. When specified (see 6.2), stowage racks shall be furnished and shall conform to the requirements specified in 3.3.1 and 3.8.1 through 3.8.4.

3.8.1 Stowage rack for class 1. Stowage rack for class 1 hose assembly shall be made of galvanized steel in accordance with ASTM A 525 with commercial thickness zinc coating or black steel in accordance with ASTM A 569.

3.8.1.1 Galvanizing. Parts of black steel shall be galvanized by the hot-dip process in accordance with ASTM A 123 with the spelter conforming to special high grade of ASTM B 6 or by electroplating with zinc in accordance with ASTM B 633. Electroplating with zinc shall be followed by a chromate treatment conforming to ASTM B 201.

3.8.2 Stowage rack for class 2. Stowage rack for class 2 hose assembly shall be made of brass or corrosion-resisting steel which has a permeability of less than 2.0 after fabrication.

3.8.3 Welding. The surfaces of all parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welding shall be in accordance with MIL-STD-278. Where zinc-coated steel is used for fabricating parts, the metallic zinc shall be removed from all joints and surfaces on which welds are to be deposited and for a distance of 1/2-inch from the expected toes of the welds. In areas where the metallic zinc cannot be removed and it is necessary to weld over the zinc-coated surfaces, electrode type 66011 in accordance with AWS A5.1 shall be used.

3.8.4 Finish. Zinc-coated surfaces damaged by shearing, punching, bending, welding, or any other fabricating process shall receive two coats of galvanizing repair paint conforming to DOD-P-21035 or a pretreatment coating conforming to DOD-P-15328 and two coats of paint conforming to MIL-E-15145.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:01Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the tests as specified in table I in the order listed (see 6.3). Failure of the hose assembly to conform to any of the requirements listed may be cause for rejection of equipment.

TABLE I. *First article tests.*

Tests	Requirement	Test
Scuff strips	3.3.8	4.5.1
Coated fabric	3.3.9.3	4.5.2
Thread	3.3.10	4.5.3
Tensile breaking strength of hose	3.4	4.5.4
Pressure drop	3.5	4.5.5
Abrasion resistance simulated service	3.6	4.5.6

4.3.1 Inspection conditions. Inspections shall be performed in accordance with the test conditions specified in 4.5.

4.3.2 First article sample. One class 1 hose assembly shall be tested as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:01Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection. Quality conformance inspection (see 6.3) shall consist of the following examination and tests:

a. Examination (see 4.4.2)
b. Heat aging (see 4.5.2)
c. Low temperature (see 4.5.2)
d. Adhesion (see 4.5.2).

4.4.1 Sampling for examination and tests.

4.4.1.1 Lot. Hose assemblies of class 1 and 2 offered for delivery at one time shall be considered a lot.

4.4.1.2 Sampling for examination. Sample hose assemblies shall be selected from each lot in accordance with table II for the examination specified in 4.4.2 (see 6.5).

TABLE II. *Sampling for examination.*

Number of lengths of hose assemblies in lot	Number of lengths of hose assemblies in sample
15 and over	7
16 to 40	10
41 to 110	15
111 to 300	25
301 to 500	35
501 and over	50

4.4.1.3 Sampling for quality conformance tests. A 1-yard sample of coated fabric shall be cut from materials used in the manufacture of the hose assemblies for the tests specified in 4.4. Samples shall be obtained in such a manner as to be representative of the hose assembly contained in the lot.

4.4.2 Examination. Each of the sample hose assemblies selected in accordance with 4.4.1.2 shall be examined and measured to verify conformance to the requirements of this specification not requiring tests. The classification of defects in table III is the minimum that shall be considered in the examination. Hose assemblies in the sample containing one or more defects shall not be delivered. Evaluation of defects through visual examination shall be in accordance with FED-STD-162.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:01Z
Check the source to verify that this is the current version before use.

TABLE III. *Classification of defects.*

Categories	Defects
Critical	None defined
Major	
101	Size, class, and material not as specified
102	Incomplete, component parts missing or improperly assembled
103	Limiting dimensions exceeded
104	Protective coatings missing, incomplete, or not as specified
105	Placing of aluminum and aluminum alloys directly with copper or copper-bearing alloys
106	Helical spring or connectors material defective or not as specified
107	Scuff strips, material defective, not nonmetallic flame resistant
108	Fabric, coated fabric and thread, material defective
109	Ground, helical wire to end connector, open
Minor	
201	Each hose length does not have marking

4.5 Tests.

4.5.1 Scuff strips. The flame resistance (vertical) test of the scuff strip shall be conducted in accordance with method 5903 of FED-STD-191. The abrasion resistance of the scuff strips shall be determined in accordance with the method 5306 of FED-STD-191.

4.5.2 Coated fabric. Tests of the coated fabric shall be conducted in accordance with the methods of FED-STD-191 specified in table IV.

4.5.3 Thread. The average breaking strength of the thread used in stitched seams shall be determined in accordance with method 4100 of the FED-STD-191.

4.5.4 Tensile breaking strength of hose. A sample of hose, at least 4 feet in length, shall be tested to failure in a power-driven tensile testing machine. Failure shall be when the hose parts to expose the stiffening wire (see 3.3.2) or to allow leakage of air from hose. The rate of travel of grip shall be 2 inches per minute.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:01Z
Check the source to verify that this is the current version before use.

TABLE IV. *Coated fabric tests.*

Test	FED-STD-191 Test method
Weight of cloth	5040
Strength	5100
Abrasion resistance (1000 gram load - CS 10 abtadet wheel)	5306
Weathering resistance (100 hours without filters)	5804
Accelerated aging, oven method (at 250 ± 2 °F for 48 hours)	5850
Blocking, high temperature effect	5872
Low temperature, effect on coated cloth (at minimum 20 ± 2 °F for 2 hours)	5874
Flame resistance of cloth (vertical)	5903

4.5.5 **Pressure drop.** Three standard 15-foot lengths of hose shall be connected together and laid out flat in such a manner as to place a 90-degree bend in the middle hose section. The 90-degree bend shall have a radius of not more than 17 inches nor less than 15 inches for 8-inch hose, 19 to 21 inches for 10-inch hose and 31 to 33 inches for 16-inch hose. The hose shall be neither stretched or compressed, but allowed to lie as it normally would in actual service with the hose on either side of the bend in straight lines. The total pressure drop of the air stream through the three sections of hose shall be measured at 500 ft^3/min flow rate. The pressure drop shall be determined as follows. Both the inlet and the outlet of the assembled hoses shall be joined to round metal ducts in such a manner as not to restrict air flow through the metal ducts or the hose. The metal ducts shall have an inside diameter equal to the nominal inside diameter of the hose. Additionally the metal duct on each end of the hose shall be provided with at least three static pressure taps mounted as described in AMCA Standard 210 from which an average static pressure can be determined. The difference between the average inlet and outlet pressure is the pressure drop. The flow rate for this test shall be measured in accordance with AMCA Standard 210 downstream of the outlet pressure taps.

4.5.6 **Abrasion resistance simulated service.** Figure 1 shows the details of an apparatus used to simulate the abrasion a hose encounters as it is raised or lowered through a hatch. The apparatus consists of a length of 3-inch steel angle over which a weighted length of hose is raised and lowered with the upper section of hose at a 63 ± 2 degree angle to the horizontal. The hose travel from fully lowered to fully raised shall be approximately 18 inches. The hose shall be raised and lowered at a nominal rate of 12 cycles per minute, with one cycle defined as the travel of the hose from fully lowered to fully raised and return to fully lowered. The weight applied to the lower end of the hose shall equal 1-1/2 time the weight of one standard 15-foot length of hose

Source: https://assist.dla.mil -- Downloaded: 202640-18T11:01Z
Check the source to verify that this is the currentversion before use.

with coupling. Guides shall be provided to ensure that the hose does not rotate about its axis during the test. The test shall be terminated when the stiffening wire is first exposed. The number of cycles to failure shall be the average of four tests performed on a single length of hose with each test conducted with the hose rotated to expose an unabraded surface to the steel angle. Additionally after completing the first two tests the length of hose shall be switched end for end and the last two tests conducted.

4.6 **Inspection of packaging.** Sample packages and packs, and the inspection of the preservation, packing, and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2 (see 6.6).

5.1 General.

5.1.1 Navy fire-retardant requirements.

5.1.1.1 Lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Level A and B	–	Type II – weather resistant Category 1 – general use
Level C	–	Type I – non-weather resistant Category 1 – general use.

5.1.1.2 Fiberboard. Unless otherwise specified (see 6.2), fiberboard used in the construction of class domestic, non-weather resistant fiberboard, and cleated fiberboard boxes, including interior packaging forms shall meet the requirements of PPP-F-320.

5.2 Preservation. Preservation shall be level A, C, or commercial as specified (see 6.2).

5.2.1 Talc/talcum. Talc/talcum when used in the packaging process of items shall be asbestos-free.

5.2.2 Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:01Z
Check the source to verify that this is the current version before use.

5.2.2.1 **Cleaning and drying.** Hose assemblies and stowage racks shall be cleaned and dried by any process or procedure of MIL-P-116 that will ensure removal of foreign materials.

5.2.2.2 **Preservative.** Preservative shall not be used.

5.2.2.3 **Unit preservation.** Each hose assembly and each stowage rack shall be individually preserved and unit protected in accordance with method III of MIL-P-116. Prior preservation, each hose assembly shall be collapsed (see 3.3.1.1) and held in position by suitable straps.

5.2.2.4 **Unit containers.** Unit containers shall conform to PPP-B-636, class weather-resistant or PPP-B-665, class 2 with selection at the option of the contractor. Container closure shall be in accordance with the applicable container specification or appendix thereto with method V closure applicable to PPP-B-636 containers.

5.2.3 **Level C.** Preservation of hose assemblies and stowage racks shall be as specified for Level A except that unit containers shall be of the non-weather resistant, domestic type of class and method I closure shall apply for PPP-B-636 containers.

5.2.4 **Commercial.** Hose assemblies and stowage racks shall be individually unit preserved in accordance with ASTM D 3951.

5.3 **Packing.** Packing when required (see 6.2), shall be level A, B, C, or commercial as specified (see 6.2).

5.3.1 **General requirements for levels A, B, and C.** Container selected (see 5.3.2) shall be a minimum weight and cube consistent with the protection required, of uniform size, and contain identical quantities.

5.3.2 **Levels A, B, and C containers.** Hose assemblies and stowage racks preserved as specified (see 5.2), shall be packed in exterior shipping containers in accordance with appendix C, table VII of MIL-STD-2073-1, for the level of packing specified (see 5.3). Unless otherwise specified (see 6.2), container selection including container options shall be at the contractor's option.

5.3.2.1 **Waterproofing.** Unless otherwise specified (see 6.2), level A and when specified (see 6.2), level B shipping containers shall be provided with caseliners, linings, wraps or shrouds in accordance with the waterproofing requirements of MIL-STD-1186.

5.3.2.2 **Closure and gross weight.**

5.3.2.2.1 **Closure.** Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with non-metallic or tape banding and domestic non-weather resistant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:01Z
Check the source to verify that this is the current version before use.

5.3.2.2.2 **Weight.** Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.3 **Commercial.** Hose assemblies and stowage racks, preserved as specified (see 5.2), shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 **Container modification.** Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two, 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage, and storage.

5.4 **Palletized unit loads.** When specified (see 6.2), containers shall be palletized in accordance with appendix F of MIL-STD-2073-1.

5.5 **Marking.**

5.5.1 **Levels A, B, C, and commercial.** In addition to any special marking required (see 6.2), interior (unit) packs, shipping containers and palletized unit loads shall be marked for shipment, stowage, and storage in accordance with appendix F of MIL-STD-2073-1, and shall include bar coding.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** Class 1 hose assemblies and stowage racks are intended for use with portable ventilating sets on ships where nonmagnetic equipment is not required. Class 2 hose assemblies and stowage racks are intended primarily for use with portable ventilating sets on minecraft and similar type ships requiring nonmagnetic equipment. This hose is intended to be used for desmoking shipboard spaces, ventilating holds, cofferdams, and unventilated storerooms prior to the entrance of personnel. This hose is not intended for ventilation spaces where hotwork is being performed.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
c. Class required (see 1.2)
d. When first article is required (see 3.1)
e. If stowage racks are required (see 3.8)
f. Navy fire-retardant requirements (see 5.1.1.1 and 5.1.1.2)

Source: https://assist.dla.mil -- Downloaded: 20260-18T11:01Z
Check the source to verify that this is the current version before use.

g. Levels of preservation and packing required (see 5.2 and 5.3)
h. When container selection is other than contractor's option (see 5.3.2)
i. When caseliners are required (see 5.3.2.1)
j. When palletization is required (see 5.4)
k. Special marking required (see 5.5.1).

6.3 **Consideration of data requirements.** The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID number	DID title	Suggested tailoring
4.3	UDI-T-23790	Report, First Article Test	–

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ____ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 **Acceptable quality levels (AQL).** Lot acceptance of sample hose assemblies selected from each lot in accordance with table II for the examination specified in 4.4.2 should be based on the acceptable criteria given in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:01Z
Check the source to verify that this is the current version before use.

TABLE V. *Acceptable quality levels.*

Number of lengths of hose assemblies in lot	Number of lengths of hose assemblies in sample	AQL = 1.5 percent		AQL = 6.5 percent	
		Acceptance number	Rejection number	Acceptance number	Rejection number
		Major defectives		Total defectives (major and minor)	
15 and over	7	0	1	1	2
16 to 40	10	0	1	1	2
41 to 110	15	0	1	2	3
111 to 300	25	1	2	3	4
301 to 500	35	1	2	5	6
501 and over	50	2	3	6	7

6.6 **Sub-contracted material and parts.** The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 **Subject term (key word) listing.**

Blower, portable
Helical spring
Hose, portable ventilation

6.8 **Changes from previous issue.** Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy – SH
(Project 4720-N036)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:01Z
Check the source to verify that this is the current version before use.

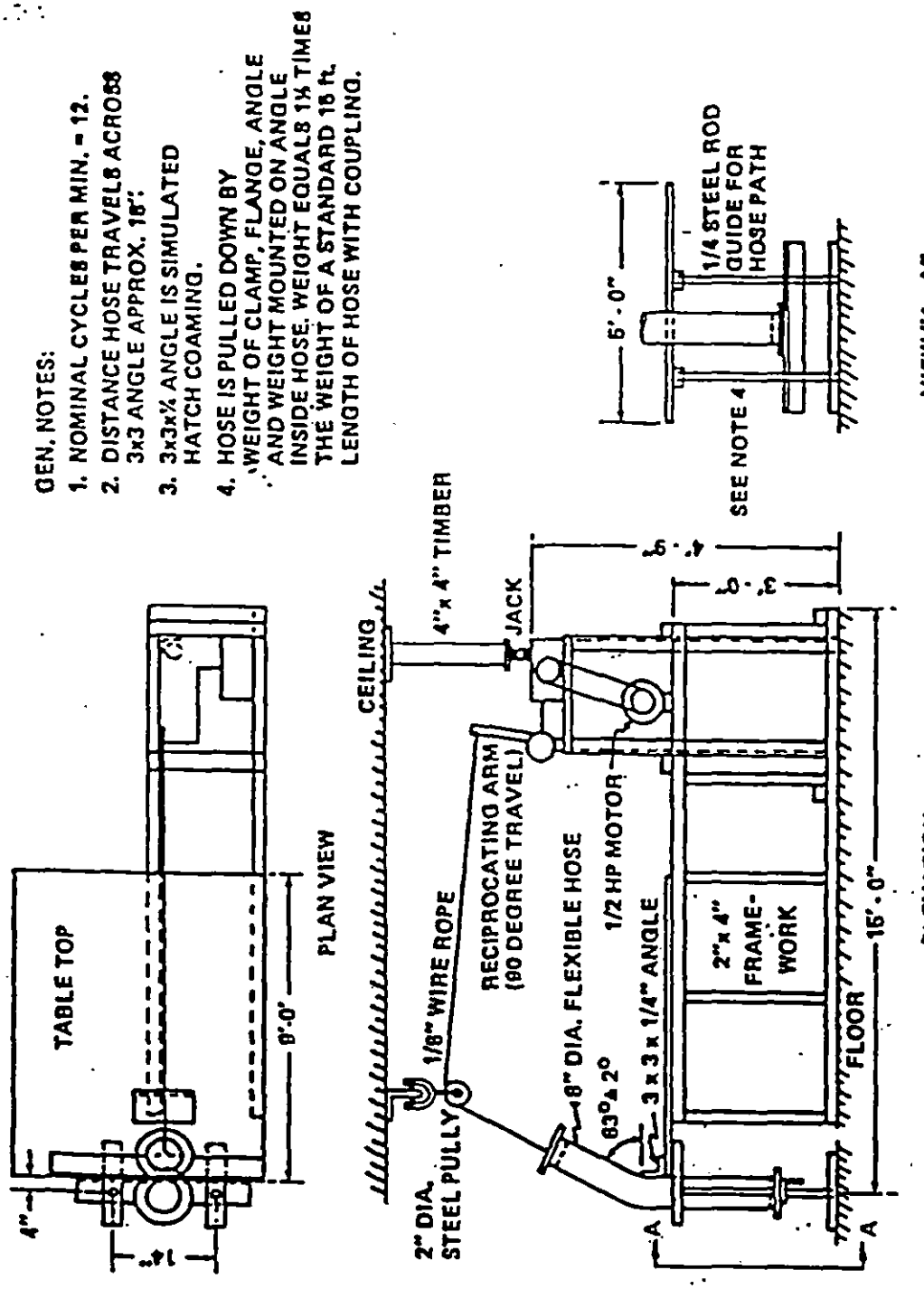


FIGURE 1. *Arrangement of apparatus for measuring abrasion resistance of flexible hose over a simulated hatch coaming.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:01Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-H-18158E(SH)	2. DOCUMENT DATE (YYMMDD) 30 November 1990

3. DOCUMENT TITLE
Hose Ducting, Low Pressure, Naval Shipboard, Portable, Ventilating Set Use

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact(TPOC): Mr. Mark Campbell (SEA 55X22) ~~PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS~~	b. TELEPHONE (Include Area Code) (1) Commercial TPOC: 703-602-7217	(2) AUTOVON 8-332-7217
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil/online/start/ Downloaded: 2022640-18T11:01Z
Check the source to verify that this is the current version before use.